EXP NO 18: Implement Perceptron based IRIS classification

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AIM

To implement Iris Flower Classification using a Perceptron-based Machine Learning algorithm and classify iris flowers into different species based on their sepal and petal measurements.

PROCEDURE

1. **Load the Iris dataset containing sepal length, sepal width, petal length, petal width, and flower species.**
2. **Preprocess the dataset and divide the data into training and testing sets.**
3. **Create and train the Perceptron model using the training data.**
4. **Predict the flower species for the test data using the trained Perceptron classifier.**
5. **Evaluate the model performance using accuracy and display the classification results.**

CODE:

```
# Car Price Prediction using Random Forest Regression


import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import LabelEncoder

from sklearn.ensemble import RandomForestRegressor

from sklearn.metrics import mean_absolute_error, r2_score


# Load dataset

df = pd.read_csv("car_data.csv")


# Encode categorical columns

le = LabelEncoder()
```

```
for col in ["Fuel_Type", "Selling_type", "Transmission"]:
    df[col] = le.fit_transform(df[col])


# Features and target
X = df[["Year", "Present_Price", "Kms_Driven",
        "Fuel_Type", "Selling_type", "Transmission"]]
y = df["Selling_Price"]


# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)


# Train model
model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X_train, y_train)


# Prediction
y_pred = model.predict(X_test)


# Evaluation
print("Mean Absolute Error:", mean_absolute_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))


# Predict price for a new car
new_car = [[2018, 8.5, 25000, 1, 1, 1]]
predicted_price = model.predict(new_car)
```

```python
print("Predicted Car Price:", round(predicted_price[0], 2), "Lakhs")# Perceptron based IRIS Classification

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import Perceptron
from sklearn.metrics import accuracy_score, confusion_matrix

# Load Iris dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create and train Perceptron
model = Perceptron(max_iter=1000, random_state=42)
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Results
print("Predicted:", y_pred)
print("Actual   :", y_test)
```

print("Accuracy :", accuracy_score(y_test, y_pred))

print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))

OUTPUT:

```
Predicted: [1 0 1 1 1 0 1 1 1 1 1 0 0 0 0 1 1 1 1 1 0 1 0 1 1 1 1 1 0 0]
Actual   : [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
Accuracy : 0.6333333333333333
Confusion Matrix:
 [[10  0  0]
 [ 0  9  0]
 [ 0 11  0]]
```

RESULT:

The Iris flower classification was successfully implemented using the Perceptron algorithm, and the model achieved good classification accuracy on the test dataset.

RESULT:

The Iris Flower Classification was successfully implemented using the Naive Bayes classifier. The model classified the three iris species—Setosa, Versicolor, and Virginica—with an accuracy of 100% on the test data.